

YASH RAWAL

+91-9399159685 | yash.mahesh.rawal@gmail.com | linkedin.com/in/rawal-yash | github.com/YashRL | Ratlam, MP, India

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GBSC Analytics & Automation Team, Mastercard

Pune, Maharashtra, India

Dear Mastercard Hiring Team,

I am writing to express my strong interest in the **AI Engineer** position within the Global Business Services Center (GBSC) Analytics and Automation team at Mastercard (Pune). With hands-on experience designing and operating enterprise GenAI platforms serving **1M+ active users** across 120+ cloud tenants, I specialize in building scalable Agentic AI workflows, low-latency RAG architectures, and Responsible AI guardrails. My technical background in cloud-native platforms (Azure, AWS), MLOps/LLMOps pipelines, and data governance directly aligns with Mastercard's mission to drive operational efficiency and secure digital innovation.

At Bigscal Technologies, I led end-to-end technical execution across system design, microservices architecture, and model deployment. Key technical achievements that match GBSC's automation priorities include:

- **Agentic AI & Workflow Automation:** Architected multi-agent routing with LangGraph & SAP Joule 2.0 discovering 650+ dynamic MCP tools. Re-engineered execution flows with capability contracts and a unified ReAct loop, delivering **3× throughput gains and \$120K+ annual compute cost savings**.
- **Enterprise RAG & Data Pipelines:** Built a 4-level hierarchical RAG ingestion and indexing pipeline over 10,000+ technical documents using GROBID and PGVector, achieving **90%+ Recall@10** retrieval accuracy.
- **Responsible AI, Security & Governance:** Architected multi-tier zero-trust PII sanitization and safety guardrails scrubbing sensitive data at tool boundaries before LLMs. Developed the Prompt Lab evaluation control plane to benchmark LLM outputs and mitigate hallucinations.
- **High-Performance Serving & Cloud Ecosystems:** Active contributor to **vLLM** (PR #55475 tool parser fix & torchcodec video decoder error boundaries). Fine-tuned and deployed Qwen2.5-7B and E5-large-v2 models across Azure and AWS infrastructure.

Operating within financial technology and global payment networks requires uncompromising standards for data ethics, confidentiality, and reliability. Having built deterministic agentic loops, strict PII redaction filters, and automated LLM-as-a-Judge benchmark suites, I am eager to help the GBSC Analytics and Automation team develop smart, cost-effective AI solutions that turn enterprise data into actionable insights.

Thank you for your time and consideration. I welcome the opportunity to discuss how my technical leadership in Generative AI, RAG optimization, and AI governance can contribute to Mastercard's ongoing success.

Sincerely,

Yash Rawal

AI/ML Engineer & Technical Lead